

☒ EASY

☐ ADVANCED MATH

Advanced Math

13

10 questions · ~15 min

Q1

ADVANCED MATH

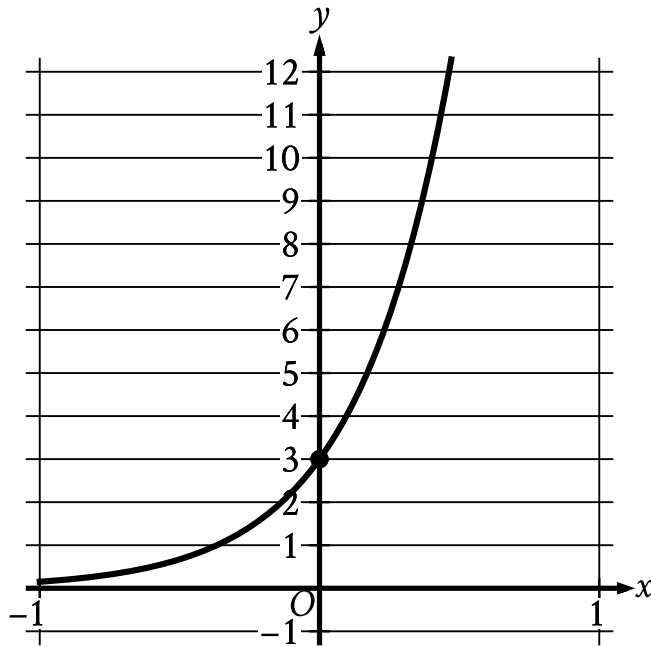
Which expression is equivalent to $50x^2 + 5x^2$?

A. $250x^2$

B. $10x^2$

C. $45x^2$

D. $55x^2$



- Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up sharply to the point (0 comma 3).
 - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
 - approximately (negative 1 comma 0.2)
 - (0 comma 3)

The graph of the exponential function f is shown, where $y = f(x)$. The y -intercept of the graph is $(0, y)$. What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0

$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$
$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$
$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$8j = k + 15m$$

The given equation relates the distinct positive numbers j , k , and m . Which equation correctly expresses j in terms of k and m ?

A. $j = \frac{k}{8} + 15m$

B. $j = k + \frac{15m}{8}$

C. $j = 8(k + 15m)$

D. $j = \frac{k + 15m}{8}$

$$f(x) = 2(3^x)$$

For the function f defined above, what is the value of $f(2)$?

- A. 9
- B. 12
- C. 18
- D. 36

$$p + 34 = q + r$$

The given equation relates the variables p , q , and r . Which equation correctly expresses p in terms of q and r ?

- A. $p = q + r + 34$
- B. $p = q + r - 34$
- C. $p = -q - r + 34$
- D. $p = -q - r - 34$

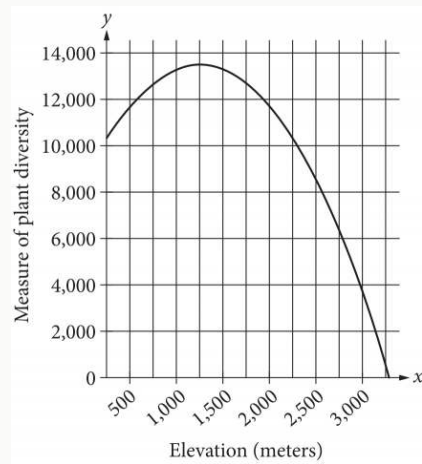
A ball is dropped from an initial height of 22 feet and bounces off the ground repeatedly. The function h estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground. Which equation defines h , where $h(n)$ is the estimated maximum height of the ball after it has hit the ground n times and n is a whole number greater than 1 and less than 10?

A. $h(n) = 22(0.22)^n$

B. $h(n) = 22(0.85)^n$

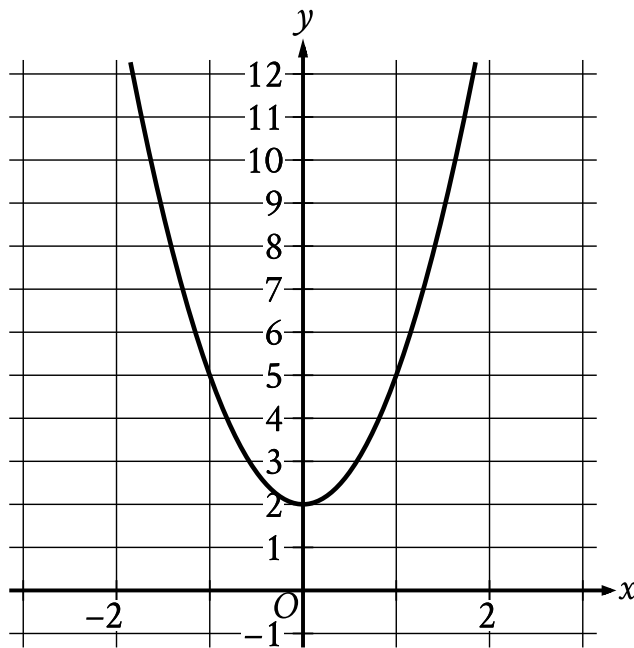
C. $h(n) = 850.22^n$

D. $h(n) = 85(0.85)^n$



The quadratic function graphed above models a particular measure of plant diversity as a function of the elevation in a region of Switzerland. According to the model, which of the following is closest to the elevation, in meters, at which plant diversity is greatest?

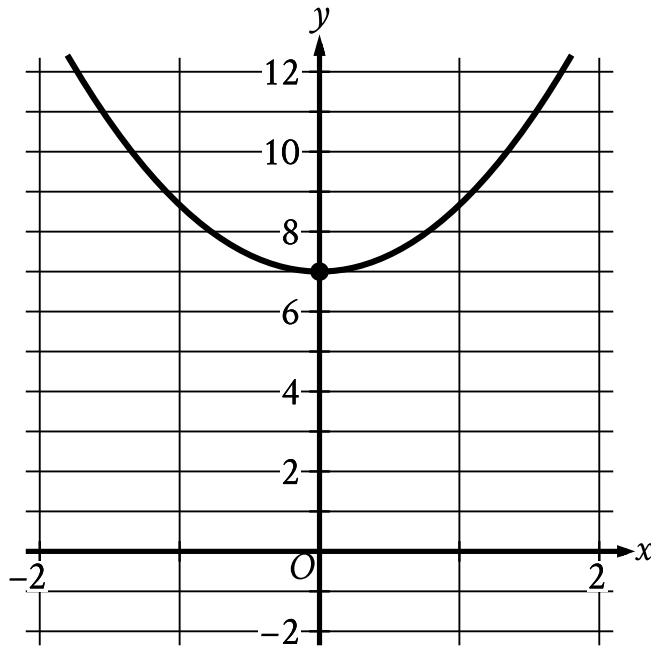
- A. 13,500
- B. 3,000
- C. 1,250
- D. 250



- The parabola opens upward.
- The parabola passes through the following points:
 - (-1, 5)
 - (0, 2)
 - (1, 5)

The graph of the quadratic function $y = f(x)$ is shown. What is the vertex of the graph?

- A. 0, -2
- B. 0, -3
- C. 0, 2
- D. 0, 3



- The parabola opens upward.
- The vertex is at the point (0 comma 7).
- The parabola passes through the following points:
 - (negative 1 comma StartFraction 26 Over 3 EndFraction)
 - (0 comma 7)
 - (1 comma StartFraction 26 Over 3 EndFraction)

The parabola shown intersects the y -axis at the point x, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
(0)	(0)	(0)	(0)
(1)	(1)	(1)	(1)
(2)	(2)	(2)	(2)
(3)	(3)	(3)	(3)
(4)	(4)	(4)	(4)

5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function g is defined by $gx = \sqrt{x} + 300$. What is the value of gx when $x = 81$?

- A. 9
- B. 300
- C. 309
- D. 381